Karl Friston - The variational foundations of movement

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eh London Institute of neurology and will be the just after his talk thank you very much that's a great pleasure to be here as some of you will know I'm slightly outside my comfort zone we've had a very very long way outside my comfort zone I found the talks this morning intriguing a lot of the rhetoric I don't understand but there's also a lot of curious similarities between many issues that we deal with in theoretical neurobiology let me move this hour his fame that I found a lot of the the rhetoric and the ideas slightly mystical that I've been hearing about but there was also a lot of formal similarities between some of the issues the wheels we deal with in theoretical neurobiology and in particular asking questions about how the brain actively infers and codes and exchanges with its environment so I think from your perspective you can regard me as the light entertainment of this workshop so what I'm going to do is try to an overview for you the basic principles that we bring to bear when trying to understand action and deception in the brain with a particular emphasis on not geometry per se but only the information geometry and the variation and the variational underpinnings of them and hopefully touched on some of the issues that we've been hearing about in the previous talk so let me overview for you what we're going to talk about I'm gonna start off in a very abstract way just asking some basic questions about the nature of self-organizing systems and in particular biological systems and I'm going to place center stage the boundary between me or an agent and the outside world with which the agent exchanges and in particular I'm gonna focus on the statistical notion of a Markov blanket that separates me the rest of the world and ask what it means to have a Markov blanket and yet still preserve a an ergodic exchange with the world that means that I can preserve certain states or I pursue certain trajectories in terms of mice of internal and external states I'm going to illustrate the nature of those principles by simulating a little primordial soup and generating a little agent within that primordial soup and performing some experiments upon it having established the basic idea and some of the principles I'm going to tell exactly the same story but from the point of view of a neuroscientist and looking at the form of the variational principles as they might be expressed in the brain and in particular looking at the notion of predictive coding and its implementation in canonical micro circuits in the brain and then I'm going to illustrate those variational principles with a few toys simulations and these are going to be very very toy from your perspective they're very simple but they are at least illustrate the basic idea I'm going to focus on action and it's observation and then if we have time conclude with a simulation of eye movements and the sampling of visual information from the world so I'm going to start again I apologize for being very abstract but I think it's quite important from my perspective at least just to motivate the underlying if I was going to be a gradient descent it could be a Gauss Newton or a gradient descent the underlying dynamic that we're going to be using so I'm going to start here well the question posed by severity and yet how can the events in space and time which take place within the spatial boundary of a living organism be accounted for by physics and chemistry now I'm not going to answer that question without what I want to highlight is a notion of a spatial boundary I've got the deeper question underlying this question how do systems induce and maintain the separation of the spatial boundary between themselves and the rest the universe and he would be the first person to acknowledge that that boundary itself is a statistical object and in statistics that would call that would come that would be basically a Markov blanket so a Markov bank is going to be a statistical boundary between some internal states of the system which I'll denote by blue here and external states in see on here that provides an installation of a statistical sort between say my States and States for the rest of the world so I'm how many people here know what a Markov blanket is good could you explain to your friends what a Markov blanket is yes basics ain't an absolutely correct okay but let me briefly unpack that for if that I'm joking I think most of you here do know after Markov blanket is so can you put your hand up if you know what a Markov processes yeah so you all know essentially what a Markov blanket is so in a with a Markov process in time the Markov blanket is just the proceeding state so it's just as it's just the states you would need to know that contain all the information about the rest of possible knowledge in all the remaining states in order to predict the current the state of reference or the internal states here so I if I wanted to predict this state given the causal influences of all other states it will be sufficient just to know the Markov blanket and I don't need to know all the remaining so these are now conditionally independent of these or vice-versa conditioned upon the Markov values and the Markov blanket comprises the parents the children and the parents of the children of the state now the moral point of view there's an interesting bipartition on that Markov blanket that is induced just by considering states that are and states that are not influenced by the internal states so for any system that can be written down in this form in terms of statistical dependences there will exist a Markov blanket and that Markov blanket can always be split into two sets which I'm going to call active states and in sensory today to know I'm going to motivate those labels just by asking you to consider that causal structure in relation to things that we know and love by our biotic or biological things from simple single cells through to say brains here so the internal States will correspond to all the internal states of the cell the active States could be the actin filaments that support the surface states of the sensory states and provide the cell with motility while the sensory states are caused by external states here that in turn influence internal states that cause change in active States that gain coupled back up to the environment and exactly the same slightly sparse causal structure can be found in the brain we have the internal states of the brain all the synaptic activities connection strengths and the like that influence and cause changes in our actuators or a texture organs that influence or changed external states that are registered by sensory states that in turn change the internal states so I'm just saying that for any system and any organism or agent or robot embedded in a larger system we can always apply this partition provided certain conditioning dependencies exist and I thought they have to exist in order to produce a distinction between me and the rest of the world through my Markov blanket and now gonna ask you should forget about that because what we're going to do is I'll just look at the generic behavior of all interesting systems and then come back and put the back of Duncan back into the equation and see what the implications are for the behavior of the sensitive states the internal states and the action states so just to motivate them so this is going to be one of a couple of slides with lots of equations or some equations on and this is the basic slide that motivates the variational approach information geometric approach to the dynamics of these systems and what I'm doing here as if in an abstract weight is taking any random dynamical system where I'm lumping together states of the world and control straight into X and having random fluctuations and I'm presupposing that we're only going to be talking about systems that have measurable characteristics that persist over time so by definition they are at Golic in a week at cents and that implies the existence of a random dynamical attractor so we have two states here and they trace out a manifold or attracting set here as time evolves with multiple trajectories and we can interpret that ensemble of trajectories or attracting set in terms of an ergodic probability the probability theis sampled the system at any point in time the probability of I would find it in that state will be denoted by the density here of these trajectories so this could be say the states of a system whilst walking for example so if this attracting set exists if the system exists then it is equipped with all it has this probabilistic description that can be described by the proper Planck equation is is this a familiar object to many people here yeah okay also known as a master equation Congre forward equation lots of different names for those people who don't know it's relatively simple I just explain what it does in a second but from our perspective it's form is not terribly important what is important is it is its solution because if this attracting set exists then the rate of change of this probability is zero which means I can rearrange the solution to express the flow of states as a function of the gradients of the log probably distribution so this is the solution that I'm going to filter it is this one solution here that I'm going to use as a variational principle to understand all internal state changes and all action open systems that exist just heuristically for those people who are not familiar with the fokker-planck formalism it's very simple all it's saying here is that if you imagine just dropping it's a drop of ink into the ocean then these random fluctuations they will disappear the concentration of being throughout the ocean so in that instance it wouldn't happen attracting said he wouldn't have behaviors and we can measure and admire or put on YouTube so it must be the case then that there is some deterministic flow that is countering the dispersive forces so it's acting in opposition to the dispersion which degrades areas of high concentration high probability the distribution or density in other words it's actually flowing towards regions of high concentration and when those two dispersive and deterministic flow components are in balance then the system becomes attains non equilibrium steady state and is we clear Datuk and that's all that we're describing here this equation here is just an expression of the Helmholtz decomposition is is this expressing this flow in terms of a hill climbing or gradient ascent on the log probability and a solenoidal or divergence free component that actually breaks the detailed balance of these of a system in general and we're going to use both terms later on in a way which I think many people will be more familiar with and this cartoon is just to show that there are two components in this flow but the essence is that the flow of any system should exist has to do a hill climbing on this log of probability so now let's go back and put the Markov blanket back into the mix and let's do it in terms of the states that constitute me my sensory states my African states and my internal States now that til climbing behavior is still true which means that it's still the case and all my internal states will try to maximize a log Golic probability and all my actions or my behaviors my acted states will also try to maximize this quantity here I'm going to associate these with perception and actions and motivate that for you in the rest of the talk but just to pause for a moment and just think about what this quantity means depending upon where you were educated or what community you come from what we're saying is that the system will appear to self-organize internally and behave in a way that it's going to maximize this quantity which is just the energy of states it thinks it should be in it likes to occupy it's just the variable states the adaptive States from this we can then derive what needs from a conceptual bridge to things like reinforcement learning optical control theory expected utility theory and the like the negative value is just the self informational surprisal or surprise so any system that exists will be trying to an information theoretic sentence trying to minimize its surprise and then that brings us to the principle of maximum mutual information minimum redundancy and the free energy principle where free energy here provides a wrapper variational bound or approximation to the surprise itself the time integral our path integral over any trajectory will because of the ergodic assumption is the entropy and that's nice because the what this means is there's a resistance to the second law its resisting an increase in entropy and of course that is the holy grail of self in organizations energetic because it's just a statement of homeostasis it's just saying that systems biological systems that reduce the tendency to disorder keep themselves within physiological balance and within very nice examples of balance terms of say the constraints on there's loss functions or cost functions now as a statistician you will also notice that this quantity or e to the surprise is the probability of sensory data given me we find preparing me as a model of my me and my environment then this is also known as model evidence it's the probability of the sensory samples conditioned upon me existing or the model existing which means that this can be interpreted this maximization of this quantity here can be interpret in terms of the Bayesian brain evidence accumulation and predictive coding and that's these two perspectives about what I want to pursue and try and unpack in terms or process theories of how the brain might work and how it might act so just to give you a very quick flavor of how the nature of the dynamics are dis described what I've done here is if simulate a little primordial soup using a hundred and twenty eight macromolecules that I've given autonomous dynamics using a Lorenz attractor so they each one has a has its own dynamics of electrochemical sort and their spatial relationships determing forces amongst the molecules with weak electrochemical attraction and strong repulsion as a inverse function of their euclidean separation and just lumping these little macromolecules together they sort of bubble away quite happily showing obtain non equilibrium so they stay quite quickly the details of the simulation of completely unimportant you get exactly the same behavior whatever you whatever certain loosely coupled in amical system you put you write down you put a little bit of random fluctuations in it will show this sort of behavior in one parameter regime the reason I've done that is because I've written down the equations of motion and I now know the physical separation and these statistical coupling between these molecules I now know that structure that adjacency matrix which means in principle I can go in there and find some internal states and their Markov blanket and I can then look at that system and then do experimental way to try to illustrate the two behaviors that I was referring to before so we're down here is it Ripley's exactly the movie that we had on the previous slide but here I've color-coded the internal states in blue here and it'll ring here with a small tail the active states in red that surround the until state support the sensory states or the surface states that are exposed to the external states in Siam yeah and this thing is riddles around flapping its its tail quite happily exchanging again in non-equilibrium steady state with the environment for those of you interested it's very easy to find these these structures you just need to identify the mark of banking making for sure constitutes I was constituted by the parents the children the parents of the children and then split it into the active and sensitive States using the look that the rules are described before so what I'm now got is a sympathetic little organism that I can now ask does actually maintain the structural function integrity of this often referred to in terms of things like order polices and do the internal states imperfect act as little Bayesian engines or infer the hidden causes of their sensory states namely active inference so just briefly to illustrate the sorts of things which we do as neuroscientist which we're basically performing brain lesions or studying people with strokes this is an illustration what happens when you perturb the system so here are the locations of those all the elements of that little organism here reported in terms of their locations over 512 seconds under normal progression and these three panels show the equivalent locations where I made my lesions to either the active States these are very modern I'm just rendering them insensitive to the electrochemical interactions coupling so this would be like paralyzing the agent the sensory states by making it blind or giving it a little stroke by leading the internal states and the key thing the obvious thing here is that whatever I do I basically kill the organism I destroy its equilibrium and indeed we can assume now that the you know the proper maintenance of its structure and function did depend upon that hill-climbing behavior that I've Illustrated in the first few slides technically this is actually known as oscillator death and it's closely related to something else general a synchronization which I'll I'll mention in a second is the second illustration here just for illustrative purposes all to make a heuristic point anyway I'm asking the question is there anything in or don't do the electrochemical states of the internal states predict the physical motion of the outside world so this would be like a brain imaging experiment where I present some moving stimuli in the outside world I look for event responses in the visual cortex of the brain and it's very easy to find mixtures here that if all the internal space lagged over time by linear mixtures that do indeed predict the motion of external states here the degree of prediction and color coded by the the this the depth of this I am coloring here's the most predictable motion of the most predictable external molecule in shown in the dotted line and the prediction is a solid line and what's happening occasionally this there this little molecule here gets expelled on the sweep and then falls back again and it seems as if this thing is actually predicting or registering these external events despite the fact that they are statistically conditionally independent they're separated by the Markov blanket there is none Sheamus examples you can almost see the fluctuation internal states the Evoque response associated with this outside motion event here if you look carefully you can see that the changes in the internal States actually precede the external states which begs an important question are the internal states causing the external states or is the external fluctuation causing and registering a representation of a statistical sort in the internal state and of course the answer is both all we're seeing here is an illustration of generalized synchrony so I'm sure you all know this but this is the phenomena where if you originally observed by Huygens where you have multiple clocks hanging from the same wall or beam and ultimately they have to come to swing in synchrony there is that is their only long-term attracting state this is one of his drawings a to illustrate two plots since when in fact the same beam and that's all we're seeing it was seeing a generalized synchrony between internal and external States just by the very existence of these random dynamic attractors and here we can think about one o'clock have air comprising the internal states the other clock the external states and their Markov blanket the beam on the wall from which they are suspended I like this example because it highlights a complete symmetry of what's going on here so if you subscribe to this idea that we are trying to predict and model our world what this means is that the world is trying to model you in an exactly symmetrical way the world is watching you in exactly the same way that you are watching the world so that's all we sort of heavy lifting done from the point of view of the self-organization to summarize and the existence of a Markov blanket necessarily implies a partition of states into internal States a Markov blanket name is central active States an external or hidden state and because active States change but not change by external states they minimize the entropy of internal states in the Markov blanket which means that action will appear to maintain the structural and functional integrity of a Markov blanket and sent our internal states will appear to infer the hidden causes of sensory states by maximizing Bayesian model evidence and influence those hidden causes through action and we will refer to that as an active imprint so now what I want to do is just repeat the exactly the same story but using a different metric and the sorts of ideas and perspectives that you'd find in psychology and neuroscience but we're going to be appealing to exactly the same ideas the only slight difference here is previously I had written down a simulation where the value functional cost functional at a surprise function was an emergent property of the differential equations here are actually going to write down the the Gothic density in terms of prior beliefs about the about the the states that agents like to occupy but the the actual simulations and solutions that I'm going to show you are basically exactly the same a differential equation hill climbing equation and if that the same code so where this sort of talk normally starts is with the Helmholtz so now we're talking about the brain and the brain as a constructive an organism that constructs predictions and explanations for its sensory inputs beautifully articulated in terms in by this quote a objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous system so the brain is providing as its spending predictions of our explanations or hypotheses for the sensory input and is updating or revising its internal hypotheses in order to account for the sensory evidence at and this of course is very closely related to the notion of perceptions hypothesis testing by Richard Gregory and has been formalized and articulated by people like Peter Dan and Geoffrey Hinton from a Bayesian perspective borrowing from variational principles in particular variational free-energy invented by richard fineman to solve the firm his particular quantum electrodynamic old problems using patented or approximations to the path integral problems so that's what we're going to this is a perspective we're going to pursue this quote is nice that in the sense that it induces its notion of a sensory impression on the nervous system which of course is very much like the sense from States being or reflecting external influences of the external States are impressing themselves on the the Markov blanket or this veil to produce the sensory impressions and if it is the case that the internal states are maximizing the MOT Bayesian model evidence they are implicitly inferring the causes in the outside world that produce that so let's see what that might look like how that might be implemented in a simple brain like organism so here's our basic hill-climbing equation again in its Helmholtz form and replacing the surprise with the free energy bound here so we've actually a great descent on the free energy here with its divergence free and curl three components so nidal components and just rewriting this equation in a form that might be more familiar to many of you it's basically a Kalman filter so the the solenoidal component now becomes a prediction in the sense that prediction can't change the model evidence because there's no new data so it doesn't change the low probabilities flows on the iso contours whereas the update does the hill climbing and that's just a minimization of the dictionary way by the precision of the prediction error multiplied by the precision of the random fluctuations so what's the dictionary we've heard about that a number of times it's basically the difference between observations and some some reference here let's assume that we have these sensory impressions and that we were trying to predict the cause of these sensory impressions given some expectation mu here and if I have a generative model of forward model given these expectations of what the sensory consequences of these expected causes were I can then produce a prediction in sensory data space of sense of space look at the difference and we'll call that the prediction error and then all we're saying is that if a system exists then you should be able to write it down so it looks as if it is always trying to minimize its prediction error both by changing its internal states which now stand in for expectations of the causes and through its action noticeable there's no imperative here to actually model the true external space it is just a statement that it is sufficient to understand the dynamics in terms of minimization of prediction error so the actual cause can be very different from the true cause so that is good in the sense that we've boiled down all the behavior to a suppression of prediction error and that provides a simple intuitive perspective on perception action in the sense we can either change our brains to make our predictions more like the sensations or we can actually change the way that were sampling the sensations to make them more like the predictions both in the service of minimizing prediction error and this provides that like the metaphor fracture and perception that I was talking about previously but if this rests upon prediction errors and clearly I have to have a model that generates predictions at this point would we start to get into some of the hard problems now so let's just think about the nature of the Jered models that would actually generate data that a humanoid robot or a human actually samples from the world and let's say you set your students the task of generating foveal sampling from a synthetic retina now to do that just to generate the data let's not worry about inferring or inverting any model just to generate the data you're going to have to know the hidden causes the what and the where of what of the causes of these sensory data so you need to know what was what is the object being visually palpated what is where are we looking what are the kinetics of that saccadic search and then we'd mix these two causes with a cascade of non linear convolution operators londonium mappings different hierarchical levels to finally produce the actual sampled data

I've just written down that deep and dynamic Garrett of a forward model here in terms of hidden states or causes and hidden States per se all subject to random fluctuations eventually generating sensory consequences right at the bottom and now if we apply our hill-climbing function clueless and then rewrite the message passing implicit in this gradient descent here using the some of the Kalman filter based formulation of this we have to get a very very similar architecture we replace the random fluctuations with prediction errors but the expectations which now replace the hidden states have exactly the same form they're just cascading down generating predictions of the expectations at lower levels right down to the bottom level now the interesting thing here from the point of view of a neuroanatomist is that we actually create forward or ascending connections or message-passing where the prediction errors are propagated up or deep into the hierarchy to update or correct our expectations so there's a reciprocal message passing with descending predictions and a counter stream or ascending prediction errors all being mixed together in a relatively simple way although there are non the outers in this because the predictions of in the pitching errors are generated by nonlinear descending influences here but a fast simple and efficient way basically to map from consequences and consequences back to causes in a way that's consistent with maximizing Bayesian model evidence so just intuitively from the point of view of again say a visual neuroscientist let's to see how that might pan out in terms of anatomy and physiology we have sensory input coming in here there are top-down predictions from the visual cortex they are compared to elaborate of addiction error that the declare is passed up to the visual hierarchy to revise expectations about the causes of that say local pattern of sensory input but these predictions or expectations themselves are in receipt of higher predictions which form a second-level prediction error that can be sent up the hierarchy to revise higher and higher more abstract hierarchical deep representations and we've heard about the importance of sort of dimension reduction of factorization and in the formation or in the use of these hierarchical models that are defined by the factorization their sparse tree structure this is where you get the dimension reduction the efficient representation explanation for sensory important we can tell exactly the same story for proprioceptive input so perceptive input from stretch receptors here in the Oakland A's system coming to the pontine nuclei in the suite of top-down predictions say from the frontal eye fields we have a prediction er that could be sent forward to revise our beliefs about the configuration of our oculomotor system and provide a Bayes optimal explanation for that but and here's the important thing these prediction errors have another way of suppressing themselves they can coupled directly back to the actuators to make the stretch correspond to the descending proprioceptive prediction so they can eliminate themselves quickly and efficiently just by coupling back to the environment so this is the action this is basically action minimizing prediction error here notice that the only thing that action can change are the prediction errors of the first level not deep in the hierarchy and what I've described there is simply a reflex are nothing more nothing less in this perspective basically the descending predictions provide the reference for the reflexes to fulfill so action is in the game again changing the sampled sensations to make them more like the predictions including the predictions about the motor plant not only in the synthesis about the motor plant as I really have the reflexes right at the bottom of the of the hierarchy so these predictions of course are not simple they are richly informed by deep hierarchical processing and gathering together of all information from all modalities so these are descending predictions that are just in the proprioceptive domain but they all internally consistent with beliefs about the world and its history that have been appeared that have benefitted from the accumulation of multimodal information so these are very rich descending predictions but their actual realization by the motor plant is trivial it's just driven by the reflex arc here so these will be basically the retina reference signals we were hearing about it earlier on so person is that before I conclude with a couple of illustrations biological agents minimize their average surprising that their entropy they do this by suppressing but you can write this down in terms of the suppression of the dictionary and that can be reduced by either changing the predictions named with deception or sensations to action and this certainly speed the predictive processing of Kalman filtering scheme that I just described entailed return message passing to optimize predictions and action with flexibly makes those predictions come true and thereby minimizes surprise so I mean it's now conclude by giving you a working through three examples and sort of build upon each other which may be and speak more to robotics and open up time yeah so wait a minute I was just about to accelerate there into hyperdrive now more relaxed ago though big deal is to partner up never loop time for discussion so the first at the first so I was trying to think how all the ideas I've been I've been hearing about this morning fit with this scheme and there are lots of points of contact but there are also points of confusion for me so I'm hoping that some that will be resolved by by the audience and often that my confusion is resolved by going back to these simple toy examples yourself evidently work so the first example it was basically how would you simulate cued region so there's some change in the environment accuse a particular movement and that movement is just wreaking to target so this is actually a trivial thing to solve from the point of view of this active infant scheme and just involves putting into the model that prior beliefs that when a target changes color there is an invisible spring or an invisible spinning that is connected from the target to the tip of the agents finger becomes very stiff from poles of the finger to the target so in the absence of any cue change that string has no stiffness but as soon as the target changes the agent believes that there is an attractive force pulling its finger towards the target now if it believes that then and that is part of its generative model then it would expect to see and feel its arm move to the target and action will reflexively fulfill that and in so doing will also fulfill the visual predictions this is basically the sort of behavior we get when the target changes from green to red this is just a schematic of the implementation by reflexes through action of these descending proprioceptive predictions here so I think the interesting point here is that the forward model that implicitly resolves the you know multiple degrees of freedom problem is very simple and it's got absolutely nothing to do with the real world the real world does not contain Springs the actual model is almost a heuristic and heuristic used in a very good sense so I think that heuristics then sometimes come along with a slightly negative connotation but in fact heuristics are your priors and a profit Bayesian inference they are at the heart of everything and as you know at the very end of the final presentation we have this sort of hierarchical decomposition almost you know appealing to he respects at the very high level constructs this I think form is exactly the same as a heart a deep hierarchical genitive model where the priors induce the call empirical processes in a hierarchical model like there are heuristics then are the right nor wrong and of course once you get the action of making them come true there they come right provided they are allowed so if you've got the right heuristics they are true they're just privates and in this instance these prior beliefs resolve the usual in motor control problem because you're actually pulling not pushing you don't have to solve an approaching problem you're solving a pulling problem by use of this particular and very simple try of heuristic or pry belief that's just part of it a very simple part of the differential equation that constantly with the hierarchical generative model so this example essentially uses a point attractor so the equations of motion that describe this pry belief or this heuristic have a fixed attracting point the next example just takes exactly the same idea but with two twists first of all we're going to make the target invisible and now it's not going to have a point attractor the location of this effective target is now going to pursue a head returning cycle and we've heard about central pattern generators I'm just writing down that a central pattern generator actually has a Volterra lock to form that produces a heavy clinic cycle that attracts the effective location of this point in Euclidean though at extra personal space to a series of unstable fixed points and again it it's likely that the notion of computing your points of contact and then working out the dynamics has a lot of similarity in the way that one chance and sequences but what we're doing here with so this one equation set a very simple equation I repeat a local terror differential equation that has a number of unstable fixed points that are visited in sequence this is just nothing more or less than a central pattern generator and then there's a mapping from the location in this abstract space to certain points in the 2d planar space that attracted the finger in exactly the same way with the point attractor attracted the finger before and by different putting these points here in an appropriate place I can simulate the writing that you've just guess being traced out there now so there's an interesting simulation from the biologist's point of view because the activity of the units that are encoding these privates and generation predictions that have been filled by action do actually show a lot of characteristics that people in neurosciences familiar with so for example if I plot the activity of one of these in terms of the position in space we get placed cell activity furthermore we get placed in activity that has Direction selectivity prefers the downstroke as personally as both of the J here and another nice thing about this simulation it actually shows that or it allows me to remind myself and remind you all this behavior has been produced by deep dynamic descending predictions in the proprioceptive domain at the same time the agent is making visual predictions and feeling the consequences of its behavior and it's well happy because those predictions are completely fulfilled however if I just reduced the gain on these descending predictions I now have a Sun replay the same visual input I now have a simulation scenario where it will be like the agent watching somebody else right and yet it can use exactly the same forward model to infer the trajectory and what is being written and I simulated that here in terms of looking at the activity of this unit here when I've precluded in the movement by switching up the gain of these descending illusions but left these in play and then exam replayed the visual input to the agent and it excites exactly the same selective responses in the same forward model so it's using this generative model that has multi modal X receptive and proprioceptive predictions in the one hand to prescribe behavior through descending proprioceptive predictions but also in other context it can prescribe the X receptacle visual contour all the consequences of another's behavior so it can also accumulate sensory evidence to infer what something else what another but was doing the course that's basically the premise of the mirror neuron system which this provides a very crude model okay finally what we did there was write down explicitly the number of the attracting locations what I want to do now is to show you the last the last simulation that asked the question well how do we in visual neuroscience identify the points of attraction the next point of attraction in this electrically itinerant sampling of our sensory space and the way that I'm going to choose those is this is the second equation here when you choose those is actually quite principled it I won't go into the arguments now but basically rests upon the following argument that if it is the case that all systems that exist in allegoric sense can be written down in terms of minimizing their variational free energy and if it is the case that the behavior of all systems is determined by the empirical prize or the priors inherent in the model that forms the basis of that free energy their heuristics what are the only necessary heuristics to explain the existence of a system that doesn't go through them off expression Airi to plus or minus infinity well it's just that they believe that they minimize the energy so if that system believes it minimizes free energy in the future then it will act to fulfill that belief and it will therefore minimize free energy and it will therefore exist so what does that mean in terms of understanding the imperatives for planned action or if you like goal directed actual and I haven't got a I won't go into this it is a very interesting sort of game if one writes out the full expression for expected free engine in the future it actually entails a lot of interesting things that subsume optimal control and K control but I'm actually taking the sort of the the interesting bits out of the lost function that make its or loss functions that you would deal with and just leave them looking at what is left what is actually left is the information gain or the epistemic value the reduction in certainty I would get if I move like this so as applied to the visual domain that simply means that we're going to the point of tractors of our eyes or ocular motor system are going to be those movements that harvest or solicit or elicit sensory samples that reduce my uncertainty about the state of the world and we can compute that so if I have this hypothesis about that this image was causing my local sensory sample so you haven't moved this circle around compute the decrease in expected free energy conditioned upon that control and score it in terms of information gain epistemic value and create a salience map I can now use the Maxima of those sailors maps to drive that itinerant searching exactly the realist rate in the previous slide here's the architecture I will go into this it's not very interesting more interesting thing here is the behavior so here the locations of this successive or tunnel autonomously selected now at acting points for successive successive fixations in a movie format these are the data bit of sample stimulated EOG s or electrical I move of recordings the visual samples at the end of each is a card and this is the interesting thing in this simulation this very simple agent have beliefs that the world could either be an upside down plays a sideways face or a vertical face and it then sampled information to resolve its uncertainty or confusion about what the state of the world was that was causing its sensory information that's now shown here in terms of the expectations for the correct explanation which was the upright face the incorrect explanations and the 97 days and confidence intervals just to show that there is a within Sicard between saltatory between Sicard reduction and transit inflation of uncertainty as this agent goes and gathers bits of information then progresses really resolves its uncertainty about what's going on beyond its Markoff blanket and clearly if any evidence violated or disconfirm desai policy would then move to the next competing hypothesis very much in like a windowless as one would in a windless competition or I'm gonna take all like situation so I'll conclude with that example it would have been nice to further example that spoke more directly to robotics but I do help on I'm sorry nor is that our expertise but I will close with a quote from Helmholtz that I think again wonderfully summarizes everything that I've just said a movement we make by which we alter the appearance of objects would be thought of as an experiment designed to test whether we would have understood correctly the in variations of the phenomena before us that is the existent the existence in definite spatial relations and with that I'd like to thank all the people whose ideas I've been talking about and of course thank you for your attention thank you very much indeed and I have a question that relates to your pointing example that you then extend it I think it's too this didn't do not really well Misaka we must address for a fixed part Photography interesting plus the sequences with the exception think about I'm just coming experience the states it's ours it's fun a whole sweater yeah oh I think community people who could come and lecture on the importance of variational free energy and approximant Bayesian inference that is exactly to finesse that problem so if I understand your problem the problem you're you're highlighting if it were the case that one could actually evaluate P of s if you could actually estimate surprise or value then in principle you could just work out the gradients with respect to control variables and with respect to internal states and and then is write down the differential equation something you know a very adaptive agent that could be and while it would add that everything every aspect including the parameters would have to minimize for energy and if you write that down that becomes associative learning Instamatic time it then becomes an atlas sectional bayesian model selection and evolutionary time but everything minimizes that log energy that locket surprised because you can't measure it and as I understand it this is exactly the problem problem the mission pieman confronted when he was trying to solve the pathetical problem to work out the probability of trajectories of small particles of little united state funds and it can't be analytically solve and it certainly can't be don't possibly every evolutionary time with some you know and see at something technique but in real time that times of hence the free energy so the free energy provides an analytic bound that is really easy to measure but it renders now your optimal dynamic no longer is it a percept phase in infants it now becomes known as approximate phase in infants so the whole point of the free energy principle as opposed to just maximizing bethe model evidence or maximizing minimizing surprise is that it'll I was exactly Paul that it provides a tractable finessing of the problem that you're talking about so everything I've talked about is an approximate Bayesian inference which lends another aspect I guess to the use of the word so the free energy which is always by definition by it gives inequality bigger than the the thing that we'd like to measure to the log of the probability itself now can be written down this form so and then it has a flavor and I feel where sudden you feel comfortable because you you know and talk about likelihood models we know and talk about prize if you're comfortable with hierarchical models or deep models you will also be comfortable the notion of parametric empirical Bayes and empirical prize so these are the things that you are forced to do when doing modeling because we cannot measure usually for nonlinear models the actual thing we want to approximate which is the probability of s so probability of s given M does not appear in this equation but what we now have to do is to write down the prize the like it explicitly and notice that the entropy term is on the end here so this conforms to James's maximum entropy principle so we're trying to maximize basically the accuracy of our model whilst at the same time maximizing the entropy or the uncertainty in our explanation it's very paradoxical but this is what leads to the generalization and the simply the simplicity this is packaging of the apartment bayesian solutions and again I'm thinking about your presentations you know that was part of that that that's a simplexity of procures and the opponent abuse of the word that the the minimization of the complexity that you would you get from including this entropy term tells you or usually makes it the case that past models with nice fast hierarchical structures with low dimensions as of its explanations for high dimensional data have much more evidence or more exactly lower free energy than models that don't have that low dimensionality so you can actually write this equation out in number different ways and and that complexity becomes a kr divergence between the post view in the prior and that the determinant of that complexity which is all part of this completely standard free energy national free energy variation or bayes ensemble learning and performing ISM you get for free when you use low dimensional hierarchical models of you later so I think there's a principle drive to getting to those good heuristics I have the low dimensionality with that hierarchical temporal structuring we've been talking about it's actually mandated by the very nature of the problem that we're trying to solve or trying to explain ok so this relationship between Bayesian inference and an optimal control that that you get out of course it's the old idea going back to cowman and in his world in the in the linear world there 1 to n just like you described but later the word world by meter and flaming in the nonlinear case and then I did something and work up and did something relating into path integrals and collectively we figured out that control is actually a much bigger problem than Bayesian inference specifically what that means is that there is a subclass of stochastic optimal control problems that map want to 1/2 Bayesian inference involving the same dynamical system but there are infinitely many other do not have a inference analog and the reason is very simple actually if you look at base true and the fokker-planck and the Kushner equation they're all linear the hamilton-jacobi bellman equation is nonlinear there is a special case where it becomes linear under of logarithmic transformation but it's not always the case so I'm wondering you must have made an assumption somewhere along the way that proved a very large class of control problems and I didn't follow also where the presented I'm guessing you started by saying that P dot is 0 you kind of assume some kind of statistical equilibrium in the world which doesn't have to be true to great questions but not the very last few words when a couple of different questions and no question I've got a lot of stuff for example a lot of the things that I showed they cannot be right I can but I was cute another pet one look at the P dot 0 is is more just a statement on the conditions that we need to comply with so I was careful to say non-equilibrium steady-state so we're not talking about equilibrium we're we're talking about systems that move and change and fluctuate so by definition so if I wanted to talk about systems that have attracting sets that have measurable characteristics over extended periods of time basically the gentec-eo dissolved so that that assumption was ready to motivate the variational formulation of the hill climbing on surprise that was not a programmatic constraint on the on the solutions that I used to illustrate the ideas it's really a more fundamental motivation for where you get that fundamental hill climbing from in the first place but it doesn't it does not entail stationarity it does not entail in either a wide sensor or a weak sense it is not incident Intel thermodynamics or any other form of equilibrium it's steady state in the sense that P dot 0 Pease dot is equal to 0 I think the the BB or the more interesting question I don't know I what clearly I haven't gone through the the formalism that would map specifying the problem in terms of loss functions and how that translates into baden inference however the simple answer to that is that all the lost functions that are implicit in the illustrations here and elsewhere are all written down as log Prize so as long as we can write down your loss function in terms of a prior preference or a prior energy in the log of the prior belief then by definition it a Bayesian inference problem so the question I think then for all nonlinear and dynamical models so that's one of the conditions for doing this setting up these dualities that the any energy cost has to be a Kerr divergence between some passive and some active dynamic but there is another important requirement which is that any noise that comes into the system must live in your control space in other words if anything happened by accident you should have been able to do it yourself and that's the thing that's actually a very strong assumption so for example if I'm a robot I'm kind of shaking due to noise but my mother could have made me shake in the same way we can test that is based in infants but if somebody came and pushed me you know that I couldn't have pushed myself all of a sudden that's not a Bayesian inference problem it has to do with the noise the structure of the noise and not just with that I mean I'm gonna have to defer to you technically you know more I have been doing this I would say that this pushing you know in the sense when we get new visual information we are being pushed informational e via sensory and of course the whole point of bit Kalman filtering and you know the generalizations of that of course this is not Kalman filtering this is this is beyond extended Kalman filtering the factors in generalized coordinates emotions and generalized Bayesian filtering but that putting is of course just from the sensory perspective exactly the problem that generalized Bayesian filtering content with and result and let me prints the question back to you how is pushing the retina with visual information any different from putting my proprioception buying you know a force to my body my question is a bit naive but if I understood well that your internal state is like the brain the state of the brain and the the external state in the outside world of the brain right but the dynamics involved all state so the attractor for example contains also the the external state the internal state in the so while the internal strengths are actually just model in the external space not themselves there again the internal States are understood as modeling the external States but not themselves I've kind of feel that the dynamic model was including all states and that the the dynamic was including Nekomata general with all states right all slaves in the outside world but not the states that are doing the modeling well does it matter your question because because I didn't see exactly in the equation that you had well why the the Markov blanket was playing in the market demarco Markov blanket assumption what are we coming into the equation I see so the leap there again I apologize for rushing over that that's actually very non-trivial it's basically this thing here isn't it yeah I think everybody would be whoops would be fairly comfortable they would after half an hour the MATLAB or a pen and paper they'd be fairly comfortable with this and it's really why is this still true this basically looks as if it has eliminated the external state remember as far as the sensory active and internal states but not the external states and yet the this lot probability depends upon the experience a so that's the clever bit that's the non-trivial aspect of this result so this result still holds true when you have to go through and actually do the integration to cure it in a non trivial way so this has the same form as the previous equation oops my point so FX is a curly greater descent on this thing here where X comprises everything this is also true but this probability distribution here depends upon the external states so that's the clever bit in the sense that it's as if the internal states and the active state or more the owner of it it's as if the internal states somehow know about the external space they know where they are in the states based on external States and yet they can't because that's beyond the Markov blanket but the very existence of non-equilibrium steady state means that this has to be true and that's basically what gives these well what if what it is what gives these systems or certainly a focus on the internal set of an over system and the Markov blanket the look and feel of adapted behavior the look and feel of self-organization homeostasis and the look and feel of a little Bayesian agent thank you very much again and my question is more naivety and how the biological neural network is implementing well I think that's an open question and as open as the best theme that one would you know adopt in the robotics context so what you're asking is what what is the the the process theory that complies with the with with the principle I mean the principle in of itself is almost uninteresting because it's tautologically true I think the hard problem is really first of all the form of the genitive models that you're dealing with and the actual scheme that is used to do the liberation the energy minimization which could be selection using basic model selection it could be it could be and I concur with the enthusiasm for gauss-newton schemes certainly all our simulations essentially use a Gauss Newton scheme why well because under the particular form of approximate Bayesian inference which is not exact given by assuming the posterior is always Gaussian you can always write it down as a in terms of minimizing petitioner when you can do that you can move from it you can move from a Newton to a Gaussian scheme with all the computational savings involved so you don't have to explicitly evaluate the curvature or the Hessian so and that rests upon the fact that you've made this for class assumption in the in defining the variable free energy so that's part of the approximation here masses of computational saving all sorts of one of the things just drop out for free under that Laplace assumption so we like we in terms of neuroscientists and people with very small computers using MATLAB like the like the predictive coding formulation or Kalman filtering applying a Kalman gain to a perpendicular and that is very understandable in terms again I haven't gotten the right graphics here but it's very understandable in terms of the sorts of message passing you see in the real brain so these actually usually thought to be equivalent superficial pyramidal cells that live about a millimeter from the cortical surface the expectations are thought to be encoded by D pyramidal cells which live about four millimeters in the deeper layers of the cortex visual cortex is exactly this hierarchical structure it has exactly the right sort of a Hammond of specificity layer specificity deep and superficial it also has the right functional asymmetries remember before I said there are non the outages in this but these are in the generation of a prediction error through the nonlinear forward model which means that these descending predictions are nonlinear whereas these are linear and driving as in the Kalman filter so these particular formulations will normally be with normally associate continuous-time representations of the post we get into sitting in society basic belief obtained or belief propagation the lack of a formal distinction is between predictive coding and generalized Bayesian filtering schemes that deal with continuous state and continuous time whereas most of the microscopic models actually look at this in the neural chemistry of personality responses they only work the state space model this discrete states you know you in state Wallen k with very very large vectors so there's a formal distinction then then you're outside the game they'll have been using protein predictive coding so you've got you've got your choice so if they do more question go to lunch now and we want to thank the speaker again back in the room at 1:45